How Infant Looking Patterns to Trivalent Events Change from Object to Person Interaction

Mischa Gushiken [1] Yuexin Li [2] Peter Gordon [1]

[1]Teachers College, Columbia University [2]New York University

Introduction

How do pre-linguistic infants parse events into the argument structures required for language?

- **Home signers** evidence: deaf children of hearing parents invent gestural communication systems with argument structure, even without linguistic input (Goldin-Meadow, 1985).
- This suggests the human conceptual system constructs event representations with argument-like structures *independent of language*.
- **Trivalent events** like *giving* require three arguments [source, theme, goal], while *hugging* requires only two [agent, patient]. Do infants distinguish these?

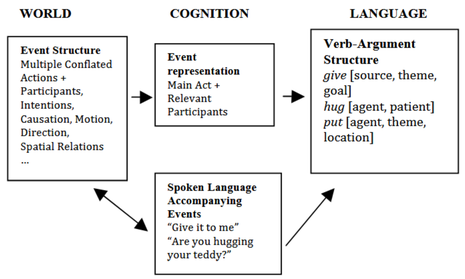

Figure 1. Mapping relations between events and language in the acquisition of verb-argument structure (Gordon, 2003).

Prior Work: Gordon (2003)

Using a **habituation paradigm** with looped videos of GIVE and HUG events:

- **10-month-olds** dishabituated when the toy was removed from GIVE but *not* from HUG, indicating the toy is relevant to GIVE's argument structure.
- **8-month-olds** showed a marginal effect (p=.07); **6-month-olds** showed no effect.
- **SHOW** (3 arguments, but more complex): even 10-month-olds failed to dishabituate, suggesting information transfer requires more advanced conceptualization.

Figure 2. Stimulus video frames from Gordon (2003). Rows show GIVE with toy, GIVE without toy, HUG with toy, and HUG without toy conditions.

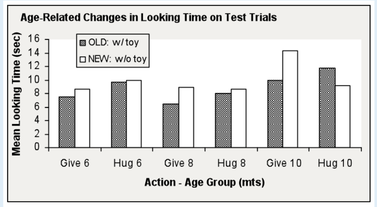

Figure 3. Age-related changes in looking time on test trials for GIVE and HUG across 6, 8, and 10 months (Gordon, 2003).

Research Questions

1. How do gaze scanning strategies shift across 7 to 11 months of age?
2. Do infants transition from tracking object movement (Toy↔Body) to attending to faces (Face↔Face)?
3. Do GIVE and SHOW conditions differ in developmental trajectory toward adult-like patterns?

Methods

Participants

40 infants (7 to 11 months) and 15 adults in a within-subjects design. Participant counts vary by condition due to data quality (GIVE: n=40; SHOW: n=45).

Stimuli & EOI Coding

Looped videos of GIVE and SHOW events. Rather than traditional screen-based AOIs, we used a custom computer-vision pipeline (**Eye-Track-ML**; Gushiken, Li, & Gordon, 2025) to automatically detect and segment event elements frame-by-frame as objects, which we call **Elements of Interest (EOIs)**.

Figure 4. GIVE stimulus frame with Eye-Track-ML SAM overlay showing segmented EOIs: man, woman, and toy.

Eye-Track-ML
Gushiken et al.,
2025

Statistical Approach

Precision-weighted Gaussian GEE with linear age trends across infant cohorts; adults as reference group. Fixation threshold: ≥3 consecutive frames; trial inclusion: ≥50% on-screen looking.

Results: GIVE Condition

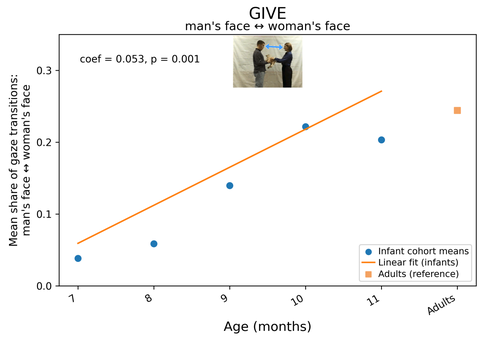

Figure 5. GIVE: Face↔Face transitions increase significantly with age (β=0.053, p=.001). By 11 months, infants reached adult levels (20.3% vs. 24.4%).

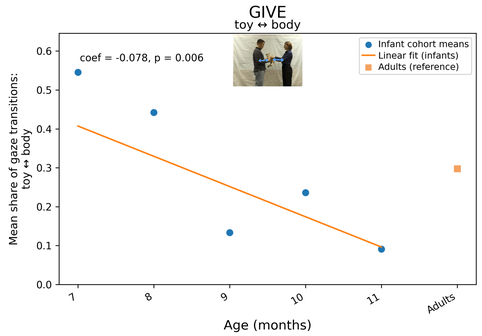

Figure 6. GIVE: Toy↔Body transitions decrease significantly with age (β=−0.078, p=.006).

Results: SHOW Condition

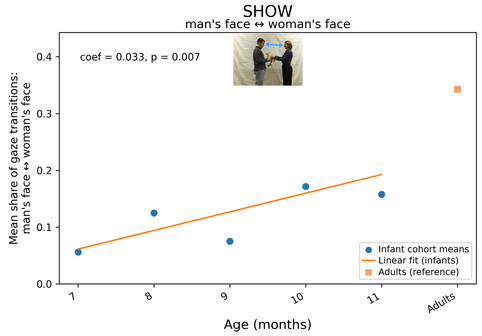

Figure 7. SHOW: Face↔Face transitions increase with age (β=0.033, p=.007), but infants *never* reach adult levels (15.8% vs. 34.3%).

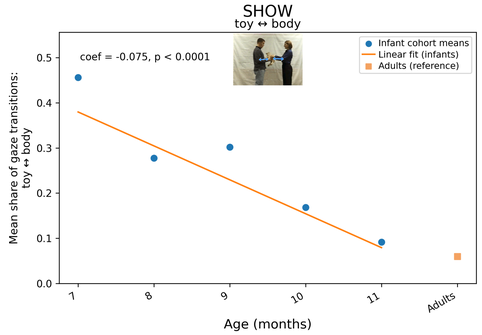

Figure 8. SHOW: Toy↔Body transitions decrease with age (β=−0.075, p<.0001).

GIVE vs. SHOW Comparison

Transition	Cond.	β	p	11-mo vs. Adult
Face↔Face	GIVE	0.053	.001	20.3% vs. 24.4%
Face↔Face	SHOW	0.033	.007	15.8% vs. 34.3%
Toy↔Body	GIVE	−0.078	.006	
Toy↔Body	SHOW	−0.075	<.0001	
Toy↔Face	GIVE	0.045	.088	marginal
Toy↔Face	SHOW	0.057	<.0001	significant

Table 1. Summary of linear age trends across GIVE and SHOW conditions.

Key insight: Infants approach adult-like gaze patterns for GIVE by 11 months (gap: 4.1%) but *not* for SHOW (gap: 18.5%), paralleling Gordon (2003) habituation findings.

Take-Away

Infants develop adult-like event parsing for GIVE by 11 months but not for SHOW, suggesting that *transfer of possession* is understood before *information transfer*. This converges with evidence from looking time (Gordon, 2003) and gaze scanning (present study). The shift from Toy↔Body to Face↔Face transitions reflects emerging understanding of **intentionality** and may scaffold verb-argument structure acquisition.

References

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